

1. Introduction

This Technical Specification describes the design, composition, processing parameters, electrical characteristics, and qualification requirements of the **MVC (Meta-Void Composite)** dielectric material developed for high-density interconnect substrates. The MVC system is intended as a direct drop-in replacement for conventional ceramic-filled PTFE and Rogers-class laminates, offering simultaneous improvements in dielectric loss, dimensional stability, and thermal management within standard PCB fabrication flows. This document supersedes all prior specifications and should be read in conjunction with *CARRIER-TO-BOARD_INTERCONNECT_DOSSIER_EN.pdf* and *MASTER_DOSSIER_ARCHITECTURE_UPGRADE_V5.0_EN.pdf*.

1.1 Scope

This specification covers MVC composite panels in laminate sheet form for multilayer PCB and substrate applications, including but not limited to: RF/mmWave front-end modules, compute package substrates, power delivery networks requiring low-loss dielectrics, and MEMS integration carriers. Dimensions, tolerances, and acceptance criteria apply to panels fabricated at qualification lots and production scale.

1.2 Applicable Documents

Ref	Document	Revision
[1]	CARRIER-TO-BOARD_INTERCONNECT_DOSSIER_EN.pdf	V5.0
[2]	MASTER_DOSSIER_ARCHITECTURE_UPGRADE_V5.0_EN.pdf	V5.0
[3]	IPC-4101E — Specification for Base Materials	Rev E
[4]	IPC-TM-650 — Test Methods Manual	Current
[5]	ASTM E1461 — Thermal Diffusivity by Flash Method	2022
[6]	IEC 62562 — Blank Detail Specification Laminates	Ed. 1

2. Material Description

The MVC composite is a two-phase heterogeneous dielectric consisting of a fluoropolymer-modified epoxy matrix reinforced with engineered hollow silica microspheres (HSM) functionalized with aminosilane coupling agents. The introduction of controlled-porosity inclusions — the 'meta-void' architecture — reduces the effective medium permittivity without sacrificing mechanical integrity.

2.1 Constituent Materials

Constituent	Grade / Supplier Class	Vol %	Function
Fluoropolymer-Modified Epoxy	FME-300 class	52 ± 2	Continuous matrix, adhesion
Hollow Silica Microspheres	HSM-G3 (D50 = 18 µm)	35 ± 3	Porosity filler, Dk reduction
E-glass woven fabric (2116 style)	IPC-4412 Class B/C	10 ± 1	Dimensional reinforcement, CTE control
Aminosilane coupling agent	APS-grade (A-1100 equiv.)	0.5–1.5	Interface adhesion promoter
Flame-retardant additive	Phosphorus-based, halogen-free	2–3	UL 94 V-0 compliance

2.2 Microstructure and Meta-Void Architecture

The meta-void architecture is defined by a bimodal pore-size distribution. Primary voids are the hollow microsphere interiors (diameter 10–25 µm); secondary voids arise from inter-sphere gaps after controlled debinding. This bimodal structure produces an effective Dk of 2.8–3.1 (10 GHz) as measured per IPC-TM-650 2.5.5.5. The connectivity of the void network is below the percolation threshold, ensuring no loss of mechanical integrity or

moisture-ingress path.

Effective Medium Approximation (Maxwell-Garnett): $\epsilon_{\text{eff}} = \epsilon_m \times (\epsilon_m + 2 \cdot \epsilon_i + 2 \cdot f \cdot (\epsilon_i - \epsilon_m)) / (\epsilon_m + 2 \cdot \epsilon_i - f \cdot (\epsilon_i - \epsilon_m))$ where f = volume fraction of inclusions, ϵ_m = matrix permittivity, ϵ_i = inclusion permittivity (≈ 1.0 for hollow voids).

3. Electrical Characteristics

3.1 Dielectric Properties vs. Frequency

Frequency	Dk (ϵ_r)	Df ($\tan \delta$)	Test Method	Condition
1 GHz	3.10 ± 0.05	0.0028 ± 0.0003	IPC-TM-650 2.5.5.5C	23 °C / 50 % RH
5 GHz	3.05 ± 0.05	0.0031 ± 0.0003	IPC-TM-650 2.5.5.5C	23 °C / 50 % RH
10 GHz	2.98 ± 0.05	0.0034 ± 0.0004	Cavity resonance	23 °C / 50 % RH
24 GHz	2.92 ± 0.06	0.0038 ± 0.0004	Split-post cavity	23 °C / 50 % RH
77 GHz	2.88 ± 0.08	0.0044 ± 0.0005	Fabry-Pérot open res.	23 °C / 50 % RH
110 GHz	2.85 ± 0.10	0.0049 ± 0.0006	Free-space method	23 °C / 50 % RH

Table 3-1: Dielectric properties measured on 1 oz Cu-clad panels, conditioned per IPC-TM-650 2.6.3.

3.2 Impedance Stability

Differential strip-line test vehicles ($50 \, \Omega \pm 1 \, \Omega$ target) fabricated on MVC panels show impedance drift of $< 0.8 \, \Omega$ over the operating temperature range $-55 \, ^\circ\text{C}$ to $+125 \, ^\circ\text{C}$, attributable to the low CTE and Dk temperature coefficient ($\text{TDk} = -8 \, \text{ppm}/^\circ\text{C}$) of the composite.

3.3 Surface and Volume Resistivity

Surface resistivity: $\geq 10^{11} \, \text{M}\Omega/\text{sq}$ (IPC-TM-650 2.5.17.1, E-24/23). Volume resistivity: $\geq 10^{11} \, \text{M}\Omega\cdot\text{cm}$ (IPC-TM-650 2.5.17.1). These values are stable after 24 h humidity conditioning (E-24/23/50).

4. Mechanical and Thermal Properties

Property	Value	Unit	Test Standard
Flexural Modulus (X-Y)	18.5 ± 1.2	GPa	IPC-TM-650 2.4.4
Flexural Strength	320 ± 25	MPa	IPC-TM-650 2.4.4
Compressive Strength	410 ± 30	MPa	ASTM D695
Tensile Modulus (X)	16.8 ± 1.0	GPa	ASTM D638
Poisson Ratio (X-Y)	0.18 ± 0.02	—	ASTM E132
Fracture Toughness KIC	0.85 ± 0.08	MPa·m ^{0.5}	ASTM E399
CTE X-Y (< Tg)	14 ± 1	ppm/°C	IPC-TM-650 2.4.41
CTE Z (< Tg)	42 ± 4	ppm/°C	IPC-TM-650 2.4.41
Glass Transition Temp Tg	185 ± 5	°C	IPC-TM-650 2.4.25
Thermal Conductivity	0.42 ± 0.04	W/m·K	ASTM E1461
Density	1.68 ± 0.03	g/cm³	ASTM D792
Water Absorption (24 h)	0.08	%	IPC-TM-650 2.6.2
Peel Strength (1 oz Cu)	≥ 1.4	N/mm	IPC-TM-650 2.4.8
Drill-ability (HAD)	< 5	µm	IPC-TM-650 2.4.42

Table 4-1: Mechanical & thermal properties (23 °C baseline unless stated).

5. Processing & Fabrication Parameters

5.1 Lamination

Parameter	Value / Range	Notes
Cure temperature	175 – 185 °C	Ramp 3 °C/min to set-point
Cure pressure	2.5 – 3.5 MPa	Hydraulic lamination press
Cure time	60 min @ set-point	After temperature stabilization
Post-cure (free-standing)	200 °C / 30 min	Optional; improves Tg +5 °C
Inner-layer bond film	MVC-BF-40 (40 µm)	Compatible prepreg system
Copper foil	RTF or VLP-2 type	Rz ≤ 1.0 µm recommended
Vacuum hold before press	≤ 5 mbar / 15 min	Outgassing step

5.2 Laser Drilling

MVC is optimised for UV (355 nm) and CO₂ (10.6 µm) laser ablation. Recommended parameters for blind via formation (diameter 75–100 µm) are given below. Absorption coefficients and ablation thresholds are tabulated in the Supplemental Material Data Sheet (PDF 2).

Laser	Wavelength	Pulse Energy	Shots / Via	Ablation Rate
UV DPSS Nd:YAG	355 nm	20 – 35 µJ	3–5	12 µm/shot
CO ₂ TEA	10.6 µm	4.0 – 6.0 mJ	8–12	8 µm/shot
Green (2ω)	532 nm	30 – 50 µJ	4–6	10 µm/shot

5.3 Chemical Compatibility & Surface Preparation

MVC exhibits broad compatibility with standard PCB wet processes. The following etchants and process chemicals are qualified:

- Copper etchants: Ammoniacal (alkaline) — fully compatible; Cupric chloride — fully compatible; Ferric chloride — compatible (limited exposure < 5 min).
- Resist developers: Na₂CO₃ (1 %) and TMAH (0.26 M) — both compatible.
- Desmear: Permanganate — standard dwell 10–12 min at 70 °C; Plasma O₂/CF₄ — compatible.
- Plating: ENIG, HASL (lead-free), Immersion Ag, Hard Au — all qualified.
- Incompatible: Concentrated H₂SO₄ > 70 %, fuming HNO₃, DMF.

6. Qualification & Acceptance Testing

All MVC lots must pass the qualification tests defined in Table 6-1. First-article inspection (FAI) is required for each new production line, new raw-material lot, and any process change affecting dielectric or mechanical properties.

Test	Standard	Accept Criterion	Frequency
Dielectric constant (Dk)	IPC-TM-650 2.5.5.5C	3.10 ± 0.05 @ 1 GHz	Every lot
Dissipation factor (Df)	IPC-TM-650 2.5.5.5C	≤ 0.003 @ 1 GHz	Every lot
Peel strength	IPC-TM-650 2.4.8	≥ 1.4 N/mm (1 oz Cu)	Every lot
Glass transition temp	IPC-TM-650 2.4.25	≥ 180 °C (DSC)	Every lot
CTE X-Y	IPC-TM-650 2.4.41	14 ± 2 ppm/°C	Quarterly
Water absorption	IPC-TM-650 2.6.2	≤ 0.10 %	Quarterly
Flammability	UL 94	V-0	Annually

IST thermal cycling	IPC-TM-650 2.6.26	< 10 % ΔR @ 500 cyc	FAI only
CAF resistance	IPC-TM-650 2.6.25	< 1 μA after 500 h	FAI only
Thermal shock	IPC-TM-650 2.6.7.2	No delamination	FAI only

Table 6-1: Qualification acceptance test matrix. FAI = First-Article Inspection.

7. Storage, Handling & Safety

7.1 Storage Conditions

- Temperature: 15 – 25 °C; avoid thermal cycling.
- Humidity: < 50 % RH; store in sealed polyethylene bags with desiccant.
- Shelf life: 12 months from manufacture date when stored per above.
- Light: protect from UV exposure; store in original packaging or light-blocking bags.
- Orientation: store panels flat; do not stack more than 20 sheets without interleaving.

7.2 Handling Precautions

Wear nitrile gloves and safety glasses when handling uncured prepreg. Avoid skin contact with uncured epoxy resin fractions. ESD precautions are not required for bare dielectric panels but must be observed once copper foils or components are attached. Do not cut MVC with conventional steel scissors — use tungsten-carbide-tipped shears or CNC routing.

7.3 Machining Dust & Waste

Machining (drilling, routing) generates respirable silica-containing dust. Use local exhaust ventilation and P100 respiratory protection. Collect waste per local silica-dust regulations. MVC swarf is non-hazardous waste (non-leaching, halogen-free) but should be segregated from metal waste streams.

8. Licensing, Traceability & Revision History

This specification and all associated design files are released as open-source hardware and software under a dual license: **AGPL-3.0** (software/firmware components) and **CERN-OHL-S v2** (hardware design files, layouts, and physical specifications). Any derivative works must be released under compatible open-source terms and must retain this license notice, the document version (V5.0), and source file references.

8.1 Source File Traceability

- TECHNICAL_SPECIFICATION.md — master specification source (this document)
- CARRIER-TO-BOARD_INTERCONNECT_DOSSIER_EN.pdf — interconnect architecture reference [1]
- MASTER_DOSSIER_ARCHITECTURE_UPGRADE_V5.0_EN.pdf — system architecture master [2]

8.2 Revision History

Rev	Date	Author	Summary of Changes
V3.0	2024-01-15	Engineering	Initial release of meta-void concept
V4.0	2024-08-22	Engineering	Added laser drilling parameters; updated CTE data
V4.2	2025-03-10	Engineering	Qualification matrix expanded; IST added
V5.0	2026-05-30	Emanuel	Full reissue: AGPL-3.0+CERN-OHL-S; bimodal void data; 110 GHz Dk data

— End of Technical Specification —